

Higgs Strahlung via contact interactions through next-to-next-to-leading order QCD

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Institute for Theoretical Particle Physics and Cosmology

Supervisor: Prof. Robert Harlander

Outline

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- Motivation and physics context

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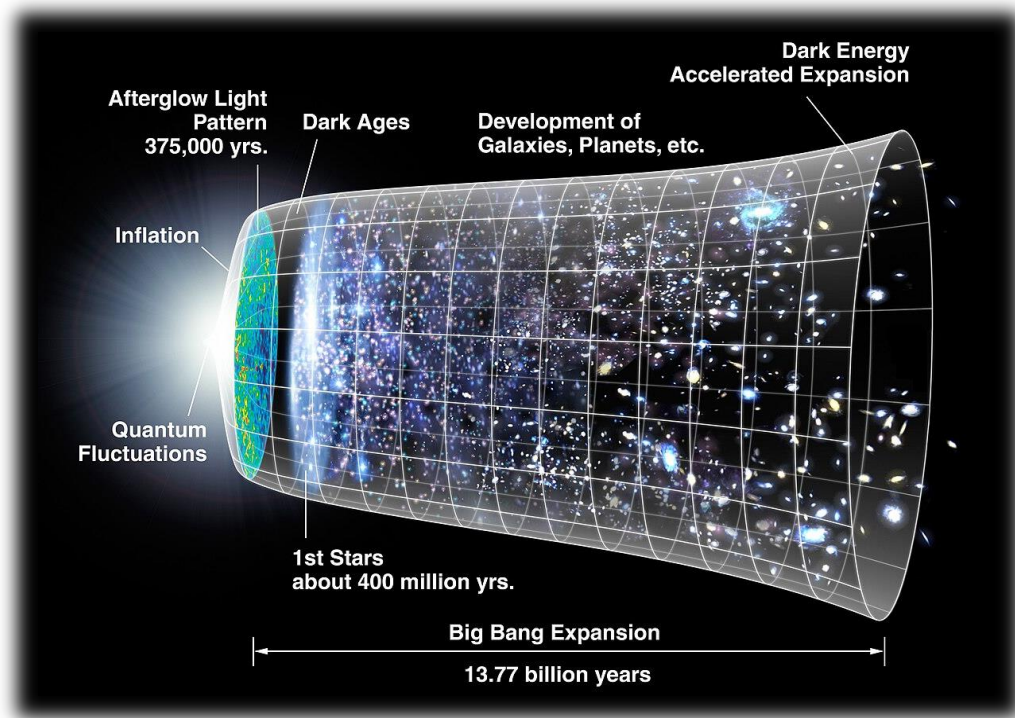
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- Demonstration of importance of the result

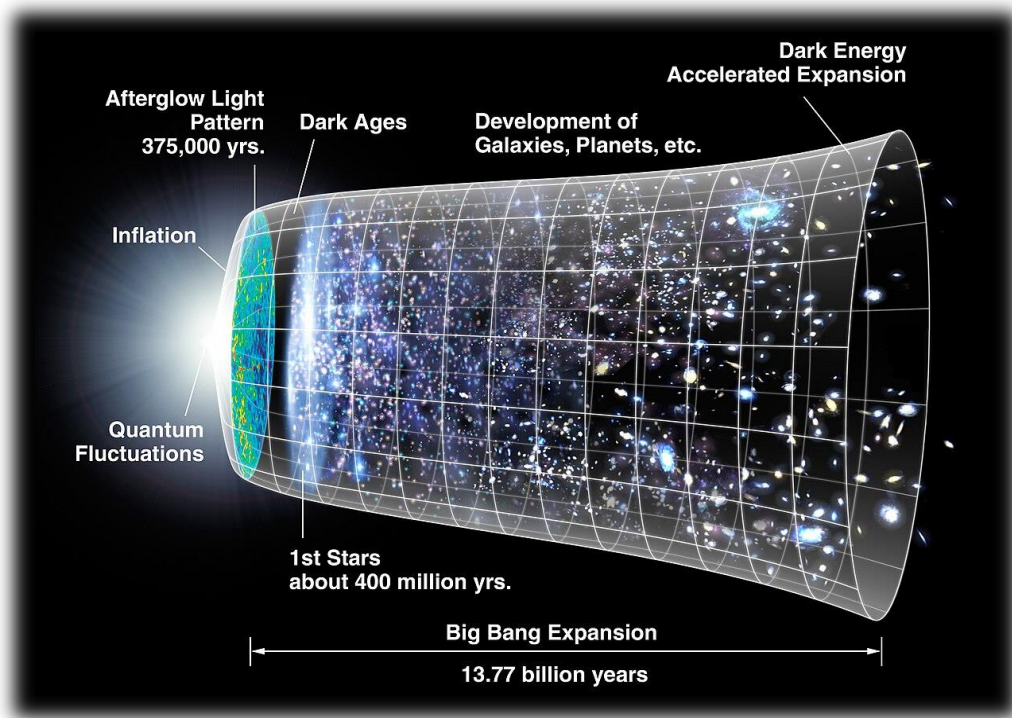
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- SMEFT framework and contact interactions
- NNLO QCD calculation
- Validation and numerical implementation
- Demonstration of importance of the result
- Summary and outlook

Understanding the Universe



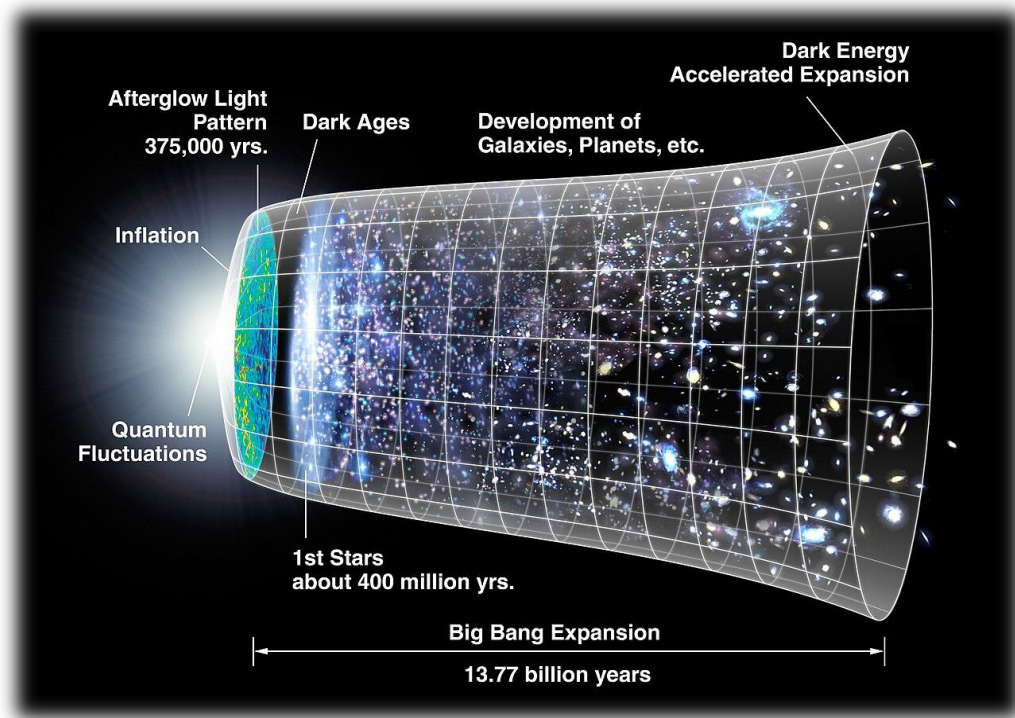
Understanding the Universe



Fundamental questions:

- What are the fundamental constituents of matter?
- What are the forces governing their interactions?
- How did the Universe evolve from the Big Bang?

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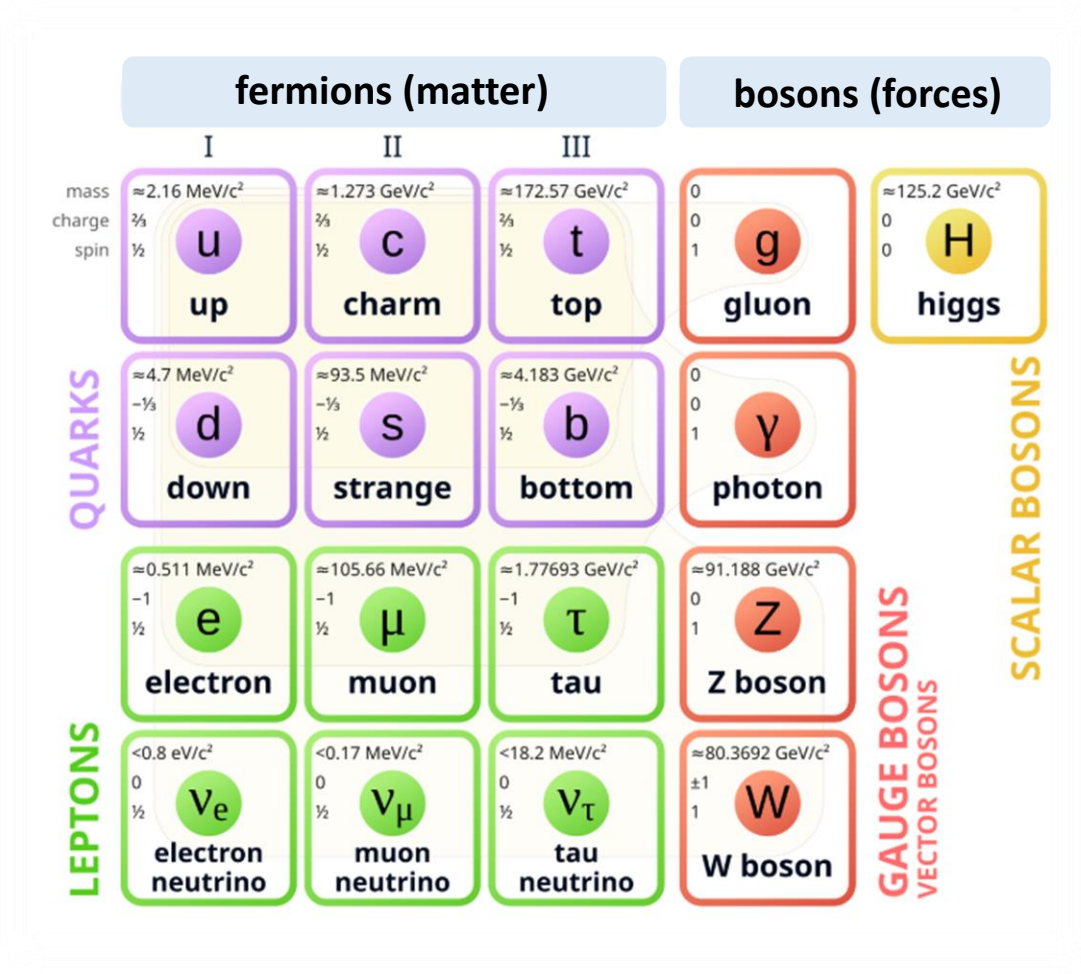
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Two complementary approaches:

- Cosmology
- Particle physics

The Standard Model of Particle Physics



The Standard Model of Particle Physics

Describes:

- Fundamental particles
- Electromagnetic, weak, and strong interactions between particles

fermions (matter)				bosons (forces)	
	I	II	III		
mass	$\approx 2.16 \text{ MeV}/c^2$	$\approx 1.273 \text{ GeV}/c^2$	$\approx 172.57 \text{ GeV}/c^2$	0	$\approx 125.2 \text{ GeV}/c^2$
charge	$\frac{2}{3}$	$\frac{2}{3}$	$\frac{2}{3}$	0	0
spin	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	1	0
QUARKS	u up	c charm	t top	g gluon	H higgs
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- Gauge bosons (force carriers)
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Extremely successful theory tested at many experiments

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The Standard Model is successful, but incomplete

- Does not explain dark matter
- Neutrinos have non-zero masses
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distance to the system $\gg$ size of the system

monopole $\rightarrow$ dipole $\rightarrow$ quadrupole $\rightarrow \dots$

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Particle Physics

Energy scale of the process $E \ll$ Energy scale of new physics Λ

$$\frac{Q^{(5)}}{\Lambda} + \frac{Q^{(6)}}{\Lambda^2} + \frac{Q^{(7)}}{\Lambda^3} + \dots$$

Standard Model Effective Field Theory

Effective Field Theory approach

- Possible new physics at scale $\Lambda \gg v$
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Standard Model Lagrangian $\mathcal{L}_{SM}$ contains

- dimension-two operator: $\varphi^\dagger \varphi$
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Why the Higgs Boson?

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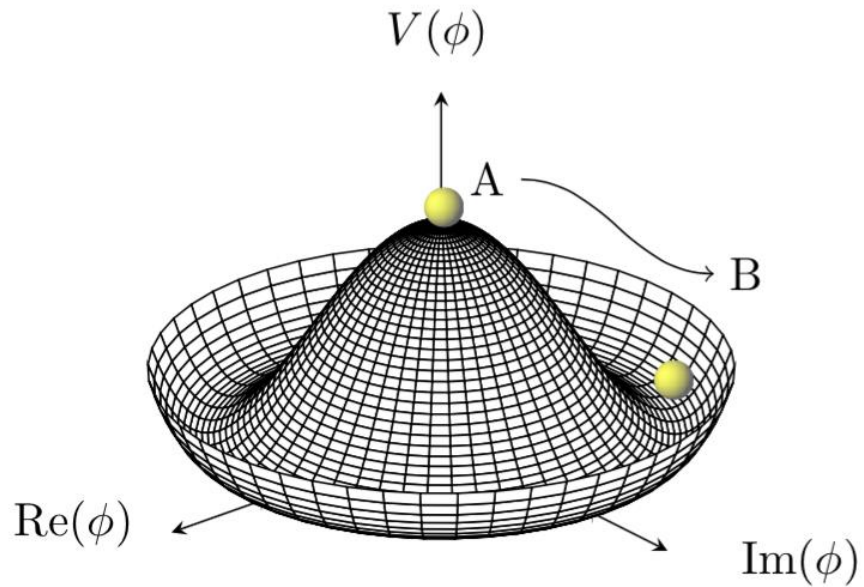
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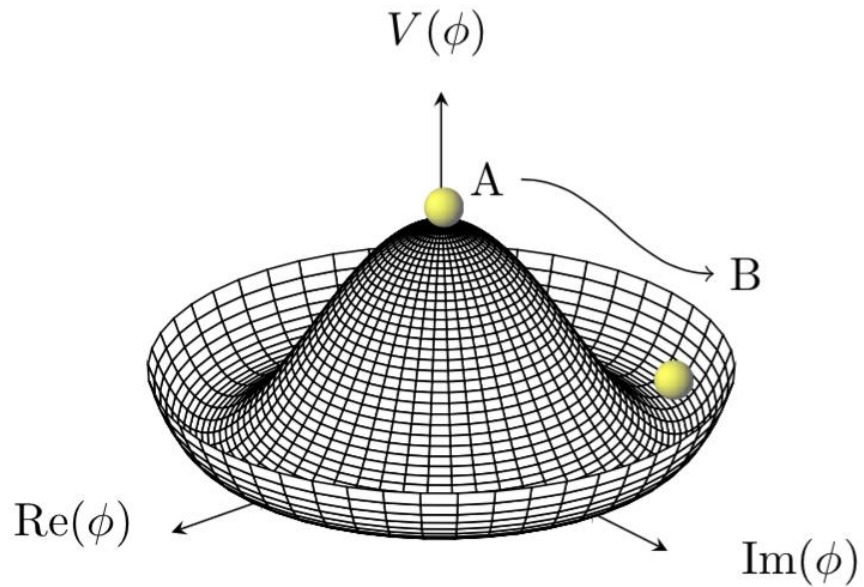


Higgs potential: spontaneous symmetry breaking

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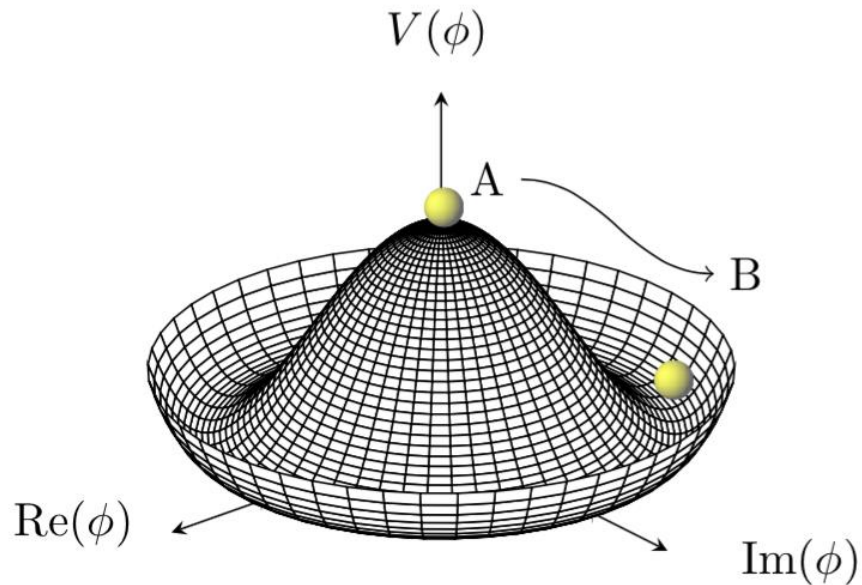


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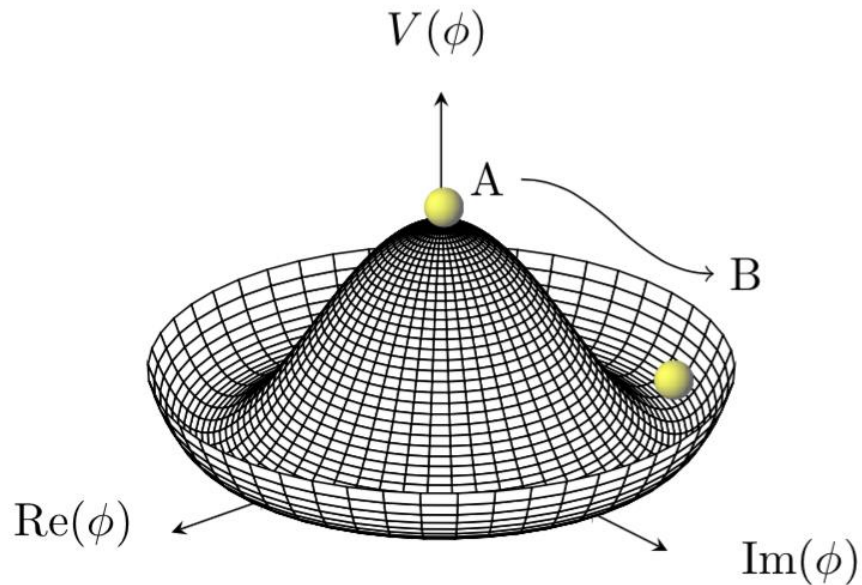


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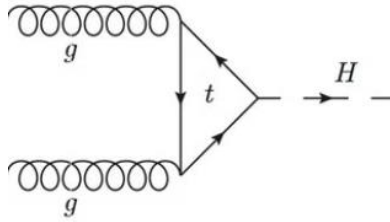
This motivates precision Higgs measurements

Higgs Production Mechanisms at the LHC

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Gluon fusion (ggF)

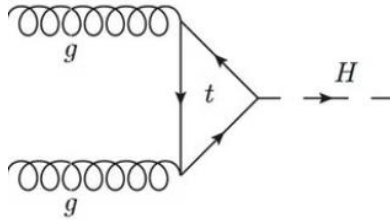
$$gg \rightarrow H$$



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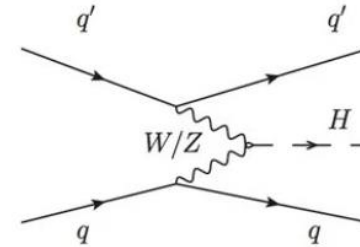
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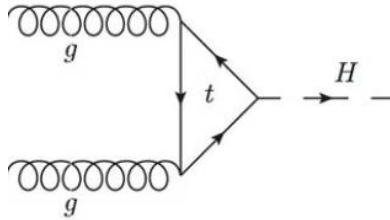
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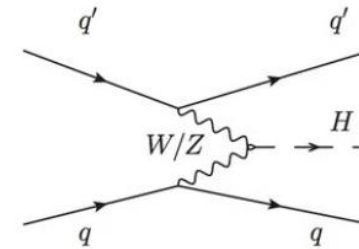
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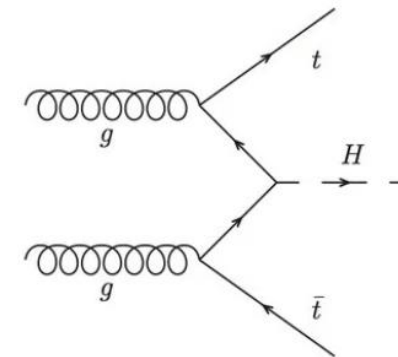
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Associated production with top quarks

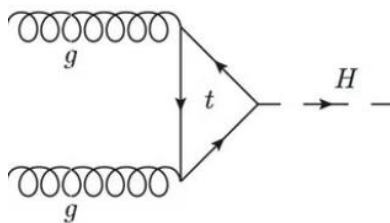
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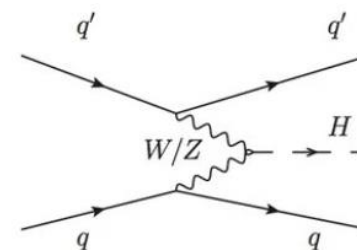
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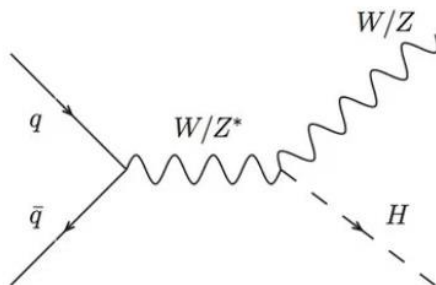
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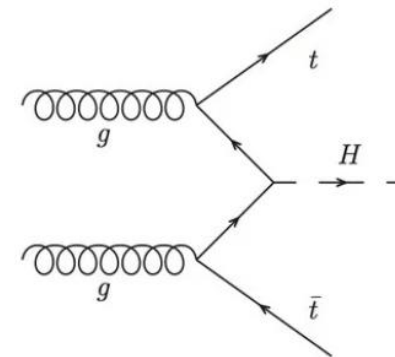
Associated production with vector boson (VH)

$$q\bar{q}' \rightarrow WH/ZH$$



Associated production with top quarks

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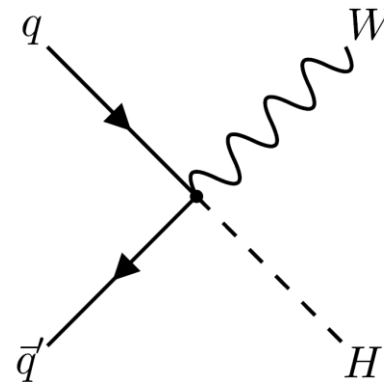
Higgs Strahlung via Contact Interactions

Dimension-six operators modify Standard Model interactions

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SMEFT contact interaction process $q\bar{q}' \rightarrow WH$

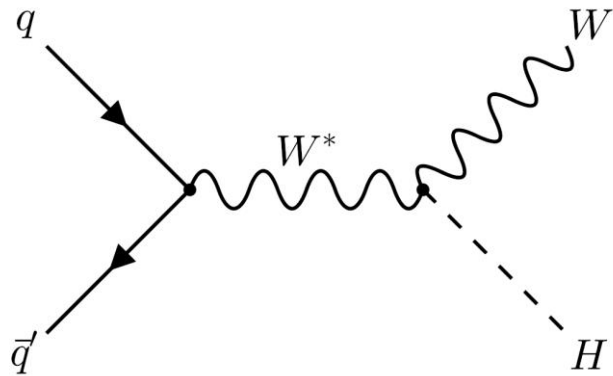


- Generated by dimension-six operators

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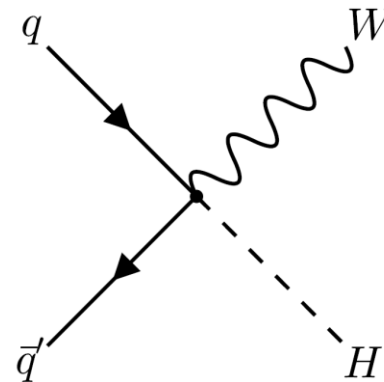
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SM Drell-Yan-like process $q\bar{q}' \rightarrow W^* \rightarrow WH$



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- Receives corrections from dimension-six operators

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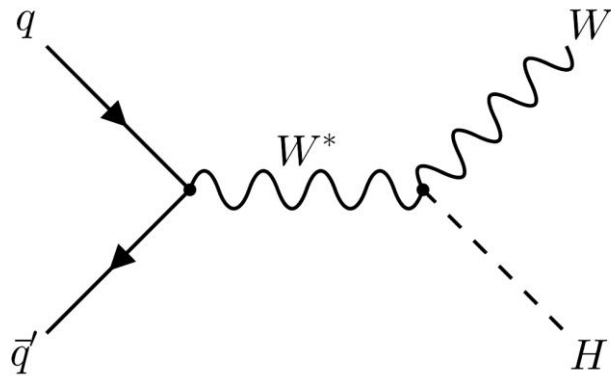


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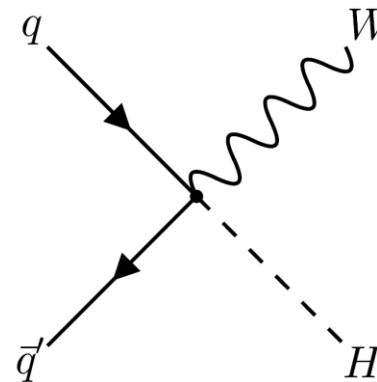
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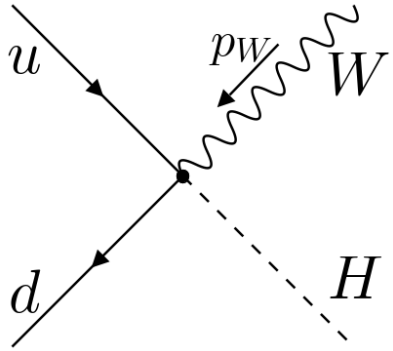


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We also consider W boson decay into a lepton-antineutrino pair $W \rightarrow l\bar{\nu}$

Higgs Strahlung via Contact Interactions

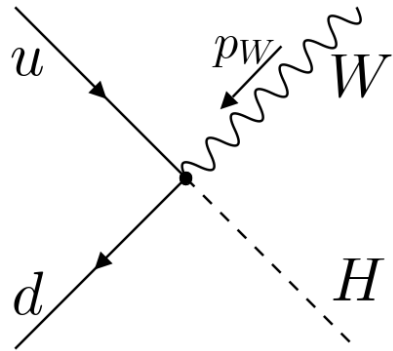
Generate contact interaction process



$$= \underline{iC_1}[\gamma^\mu, \not{p}_W]P_L + \underline{iC_2}[\gamma^\mu, \not{p}_W]P_R + \underline{iC_3}\gamma^\mu P_L + \underline{iC_4}\gamma^\mu P_R$$

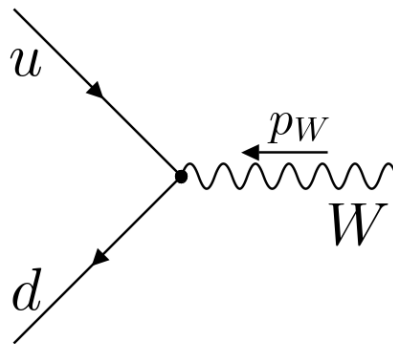
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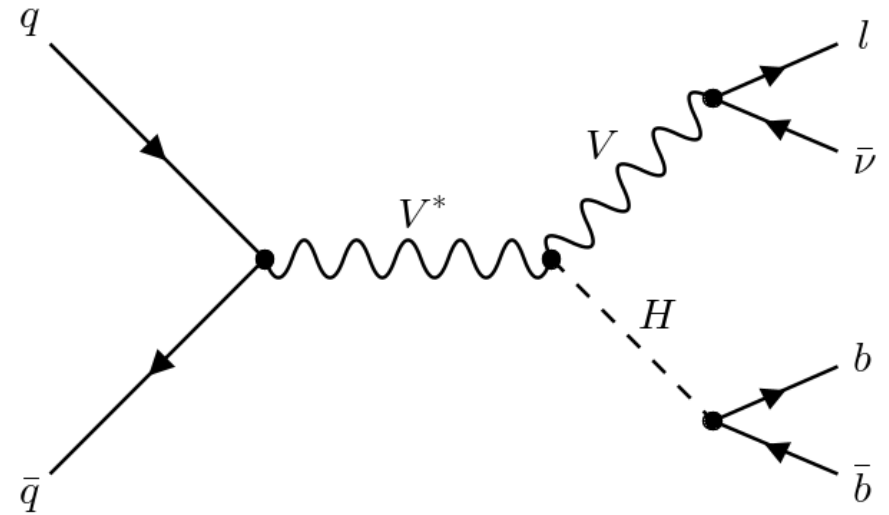
Corrections to the Standard Model process



$$= i\underline{S_3}\gamma^\mu P_L \quad \Rightarrow \quad ivC_1[\gamma^\mu, \not{p}_W]P_L + ivC_2[\gamma^\mu, \not{p}_W]P_R \\ + i\left(\underline{S_3} + \frac{v}{2}C_3\right)\gamma^\mu P_L + \frac{iv}{2}C_4\gamma^\mu P_R$$

Problem Statement

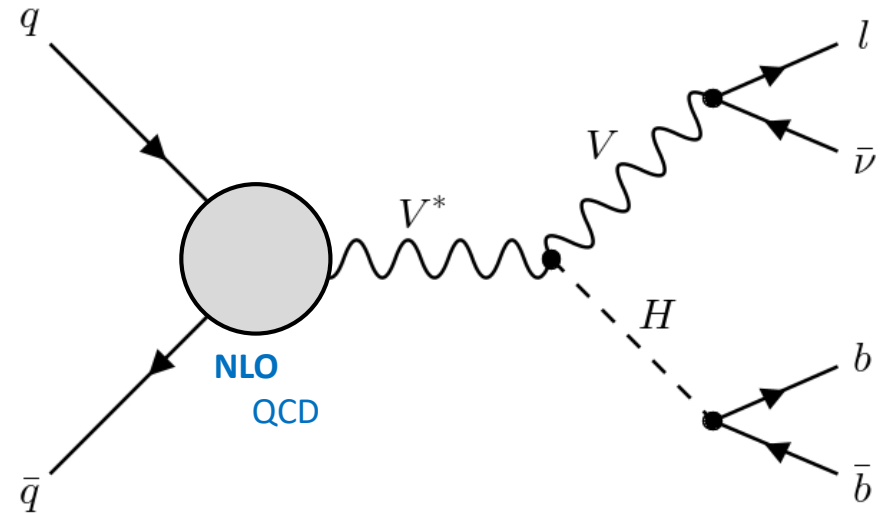
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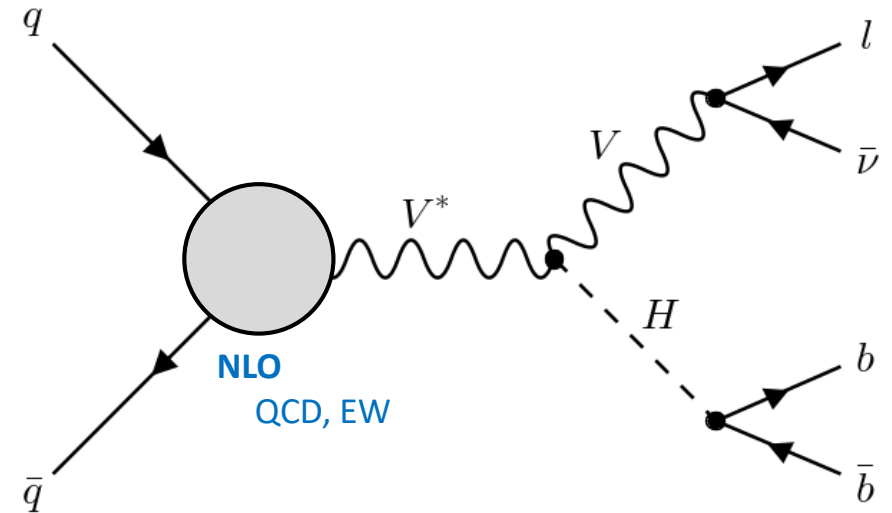
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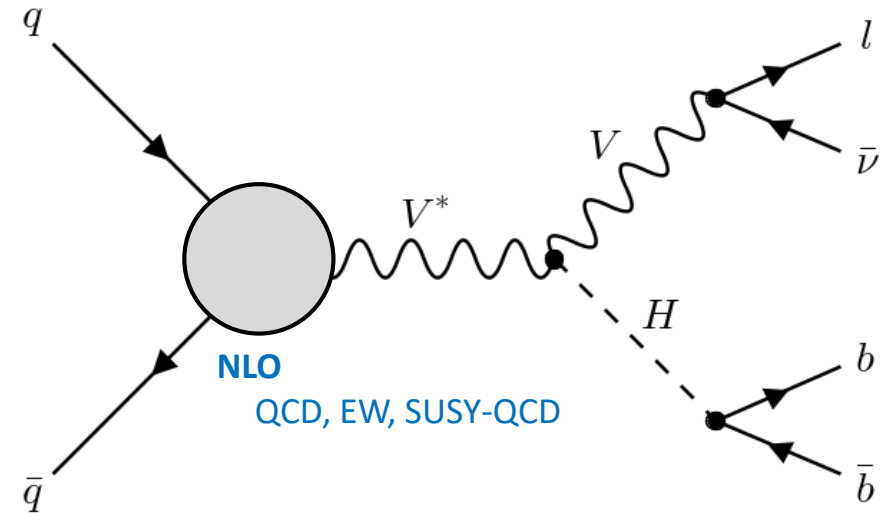
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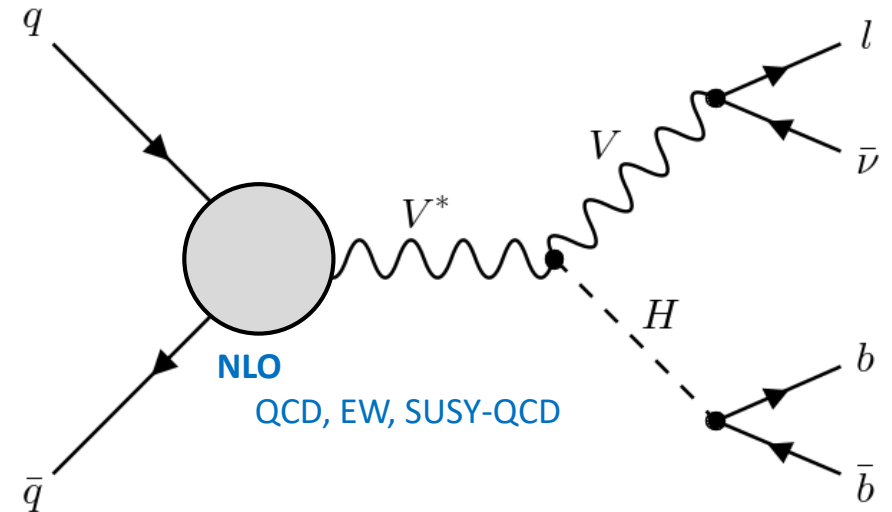
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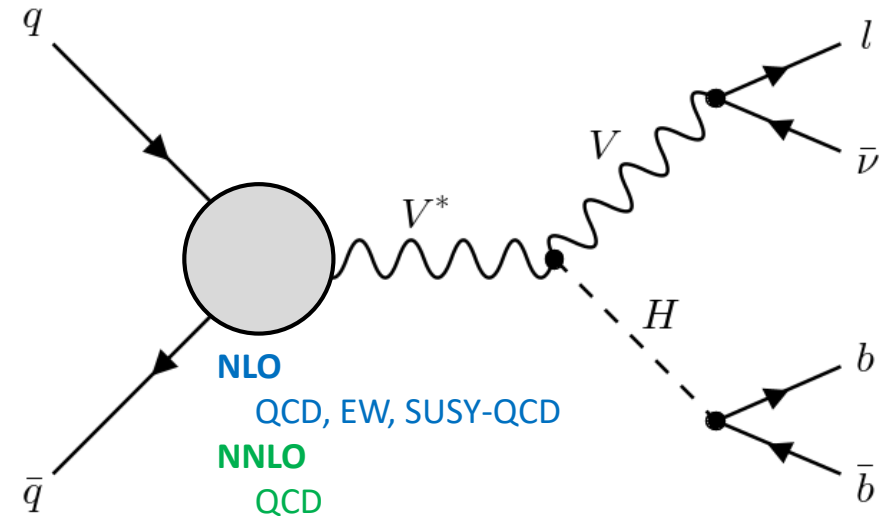
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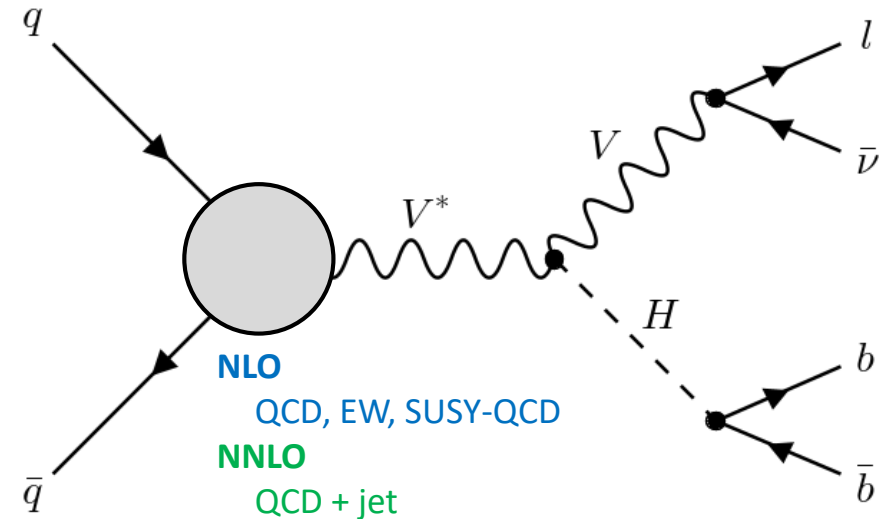
Brein et al., Phys. Lett. B 579 (2004)



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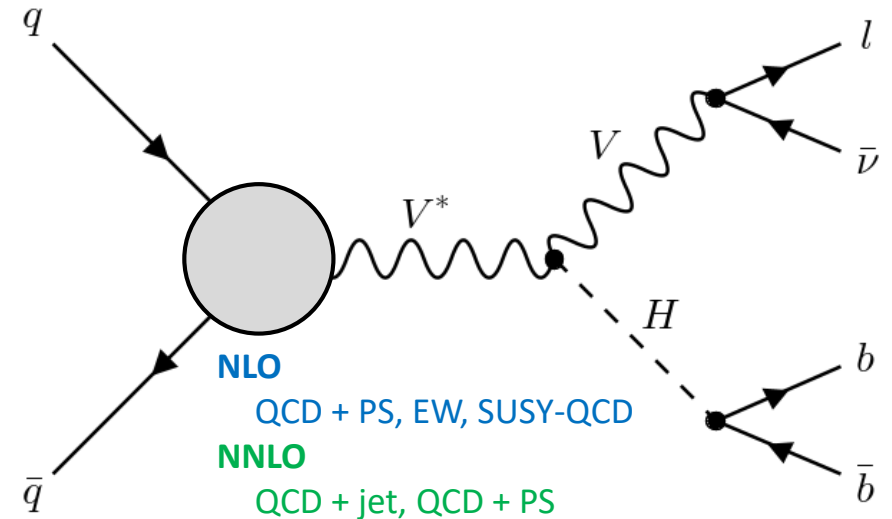
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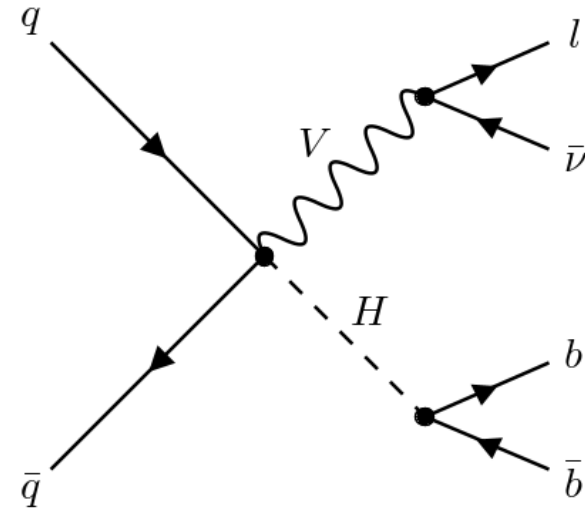
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Astill et al., JHEP 11 (2018)

Bizon et al., JHEP 06 (2020)

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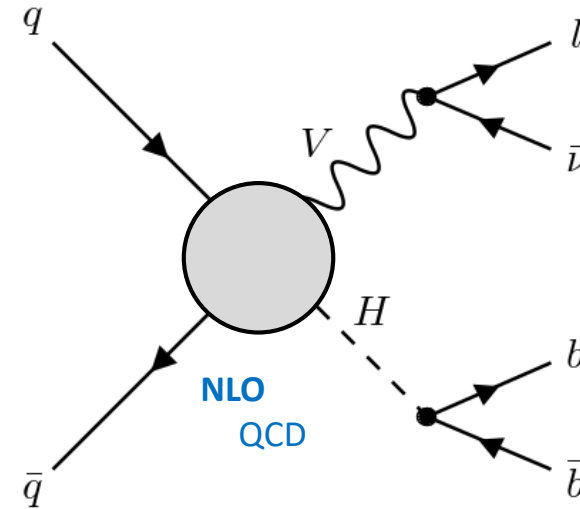
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Demartin et al., Eur. Phys. J. C 74 (2014)

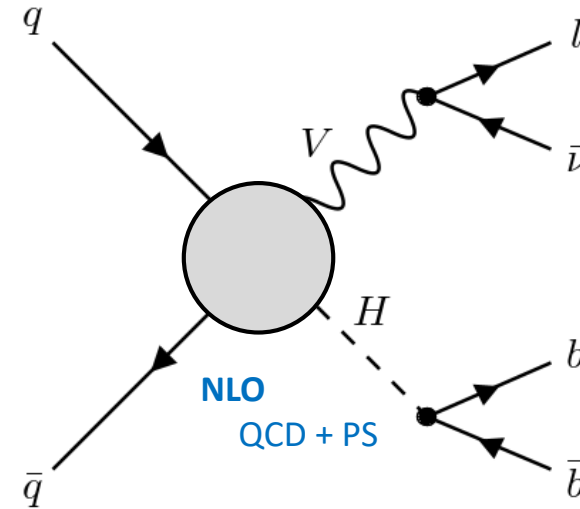
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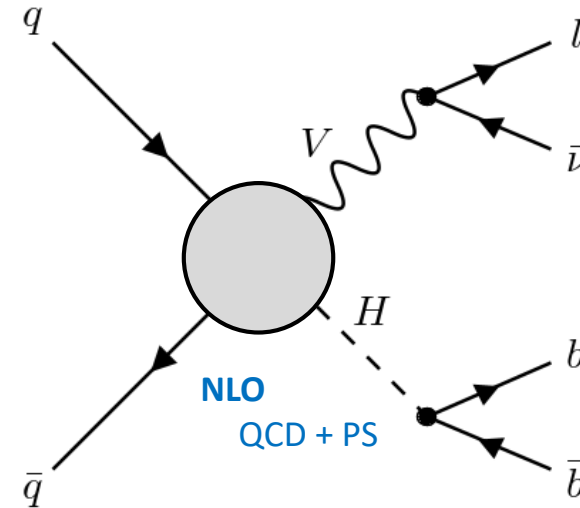
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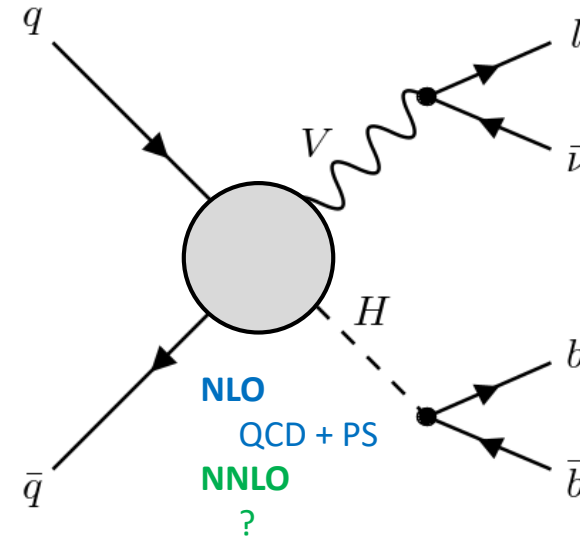
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Problem Statement

Higgs Strahlung within SMEFT: **what is done?**

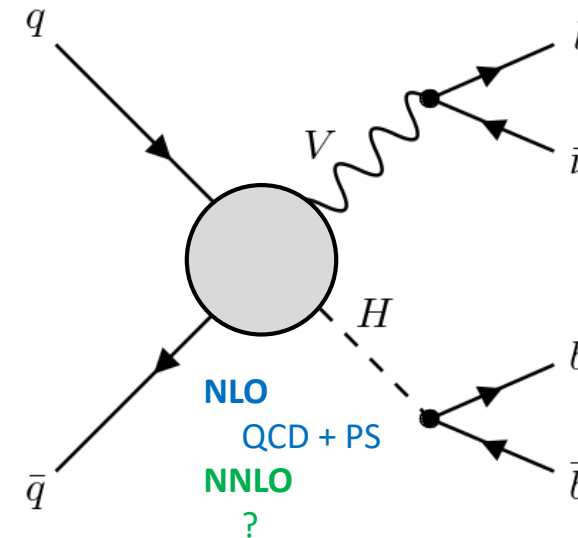
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Goal:

Compute NNLO QCD corrections to Higgs Strahlung process given by dimension-six operators within SMEFT followed by electroweak boson decay into a lepton-antineutrino pair

Perturbative QCD Corrections

Cross section expressed as series in the strong coupling $\sigma = \sigma_{LO} + \alpha_s \sigma_{NLO} + \alpha_s^2 \sigma_{NNLO} + \dots$

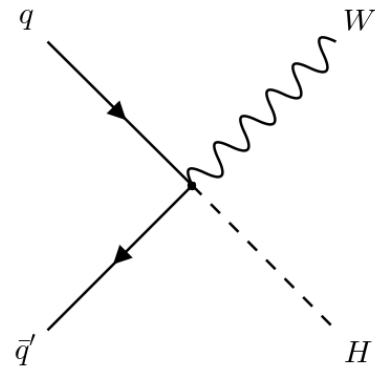
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Corrections include:

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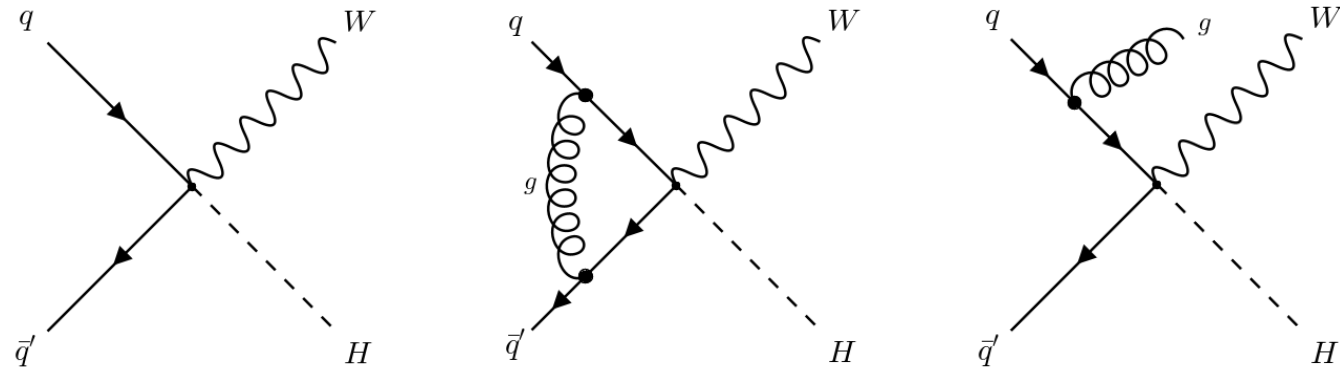
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NLO:

- virtual loop corrections
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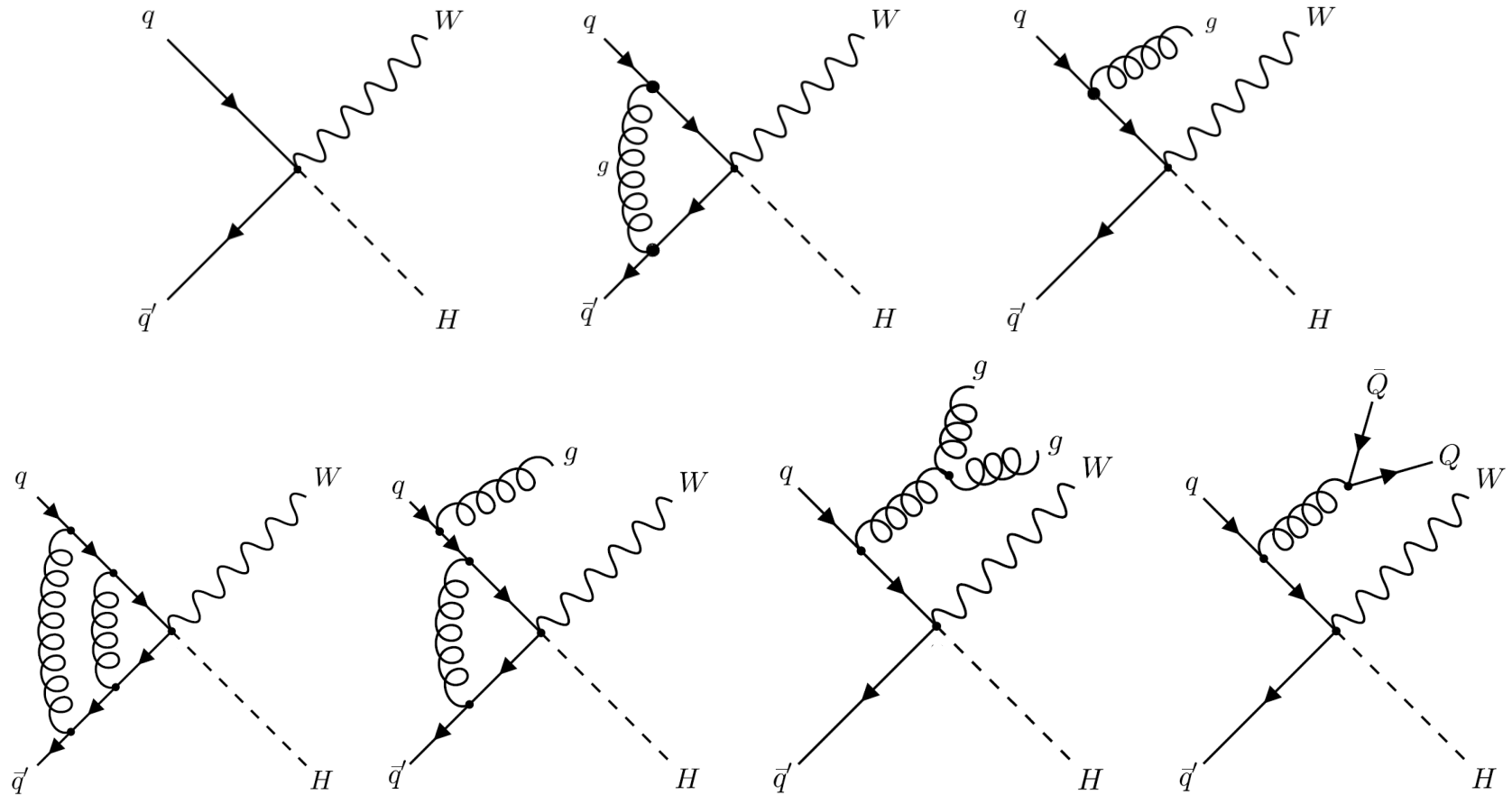
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NLO:

- virtual loop corrections
- real gluon emission

NNLO:

- two-loop diagrams
- real-virtual corrections
- double real radiation



Analytic NNLO Calculation of the Squared Amplitudes

Helicity amplitude formalism

- Decomposition of amplitudes into helicity configurations

$$\mathcal{M} = \mathcal{M}_{LRLL} + \mathcal{M}_{RLLL} + \mathcal{M}_{LLLL} + \underbrace{\mathcal{M}_{RRLL}}$$

Helicity amplitudes of
external particles $(q, \bar{q}, l, \bar{\nu})$

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- Interference term $\Rightarrow$ leading sensitivity to new physics

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Consistency Checks of the Calculation

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Comparison SM limit vs known results $\mathcal{M}_{\text{SMEFT}} \xrightarrow{C_i \rightarrow 0} \mathcal{M}_{\text{SM}}$ ✓

Numerical Validation and Phenomenological Relevance

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Input: Energy scale $\sqrt{s}$ + Wilson coefficients C_i



Phase space (external momenta)



Helicities $[p_i p_j]$ and $\langle p_i p_j \rangle$



Amplitudes and form factors



Output:

Squared amplitudes

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Amplitudes and form factors



Output: Squared amplitudes

Validation:

- Comparison with SM tools (MadGraph and OpenLoops)
- Excellent agreement
MadGraph / our result = 0.99999974
OpenLoops / our result = 0.99999971
- Non-trivial cross-check

Independent numerical validation
of analytic NNLO results

Differential Cross Sections in SMEFT

Basis decomposition

$$d\sigma = d\sigma_{SM} + C_{uW}^2 d\sigma_{uW} + C_{dW}^2 d\sigma_{dW} + C_{\varphi q(3)} d\sigma_I + C_{\varphi q(3)}^2 d\sigma_{\varphi q(3)} + C_{\varphi ud}^2 d\sigma_{\varphi ud}$$

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Key idea

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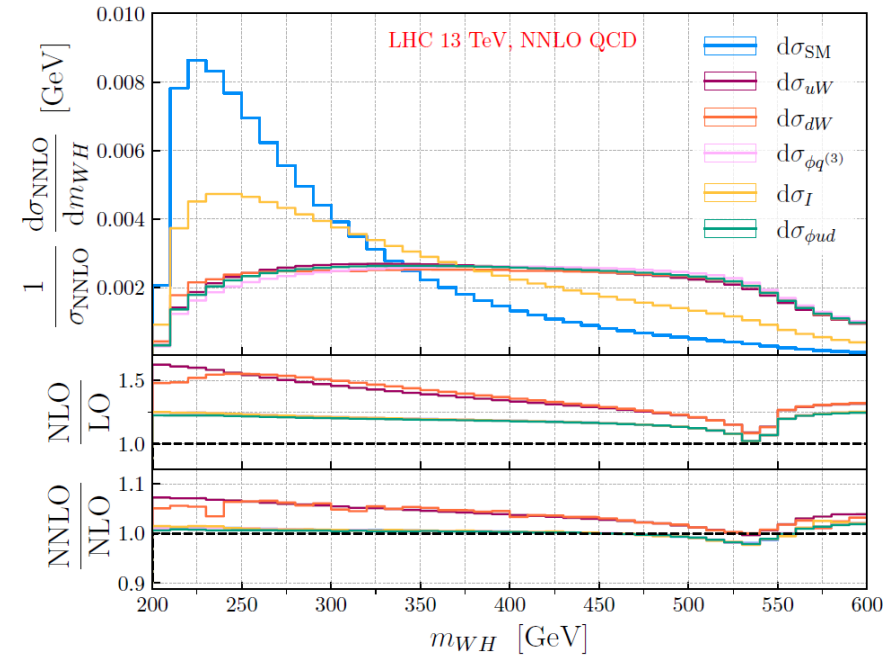
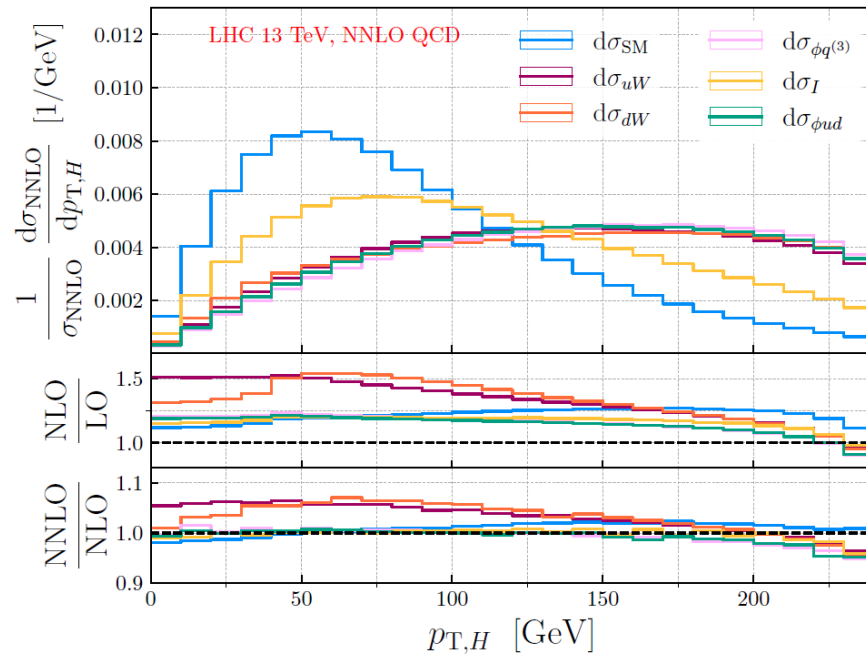
Computed once $\Rightarrow$ reused for arbitrary Wilson coefficients

Sensitivity of Kinematic Observables to SMEFT Effects

[M. Bonetti, R.V. Harlander, D. Korneev, M.-M. Long, K. Melnikov,
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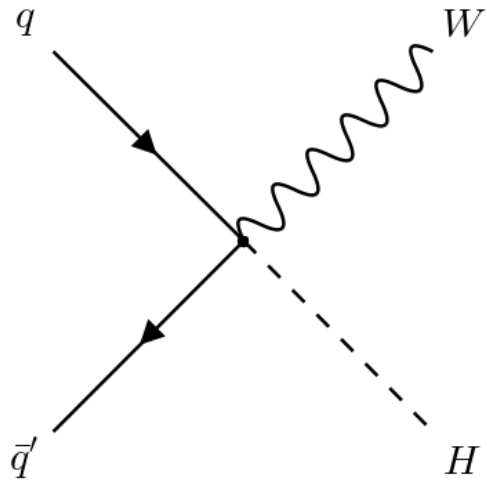


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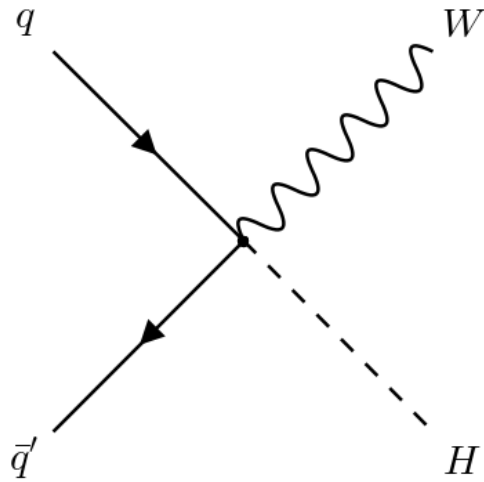
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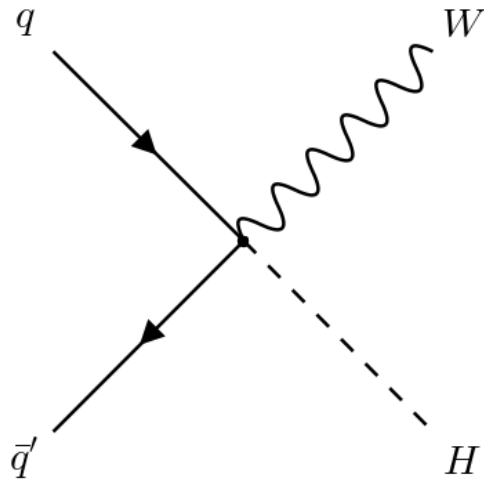


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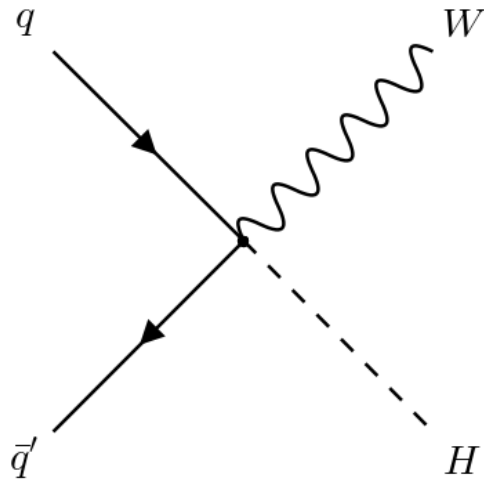


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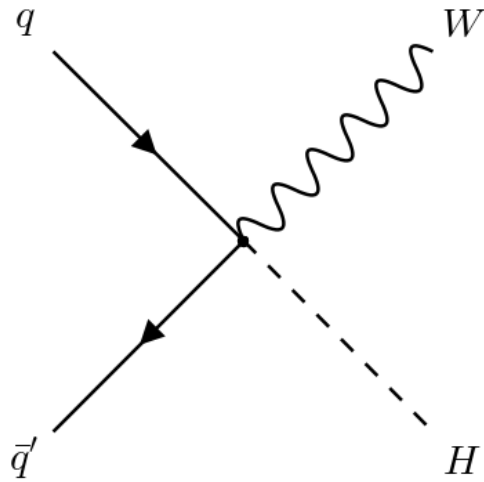


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- Analytic derivation of squared amplitudes using helicity amplitude formalism
- Validation through analytic limits and numerical cross-checks
- Results provide input for precision SMEFT studies of Higgs production at the LHC

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- Research of SMEFT behavior at high energies
- . . .

Thank you for your attention!